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Paleomagnetic evidence for a very large counterclockwise rotation of the Madre de Dios Archipelago, southern Chile

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Abstract

A paleomagnetic study of Late Paleozoic exotic units exposed in the Madre de Dios Archipelago and Diego de Almagro island, southern Chile, is reported. These units correspond to the Tarlton Limestone and the metabasaltic pillow lavas of the Denaro Complex. Nineteen sites were drilled (14 in limestone and five in lavas). Standard demagnetization procedures show that the primary remanence is not present in the investigated rocks, preventing the determination of the original latitudinal position of these rocks. However, a post-tectonic secondary magnetic component was isolated in eight sites at the Madre de Dios Archipelago, in both limestone and basaltic pillow lavas. Rock magnetic experiments and microscopic observations indicated that the carrier of this remanence is secondary titanomagnetite in the limestone and titanomagnetite–titanomaghemite in the lavas. The most likely event to produce this secondary magnetization is the nearby intrusion of the south Patagonian batholith in the Early Cretaceous. Mean direction from eight sites shows a very large ($117.7 \pm 29.9°$) counterclockwise rotation of the study area with negligible paleolatitude anomaly. This rotation is only constrained as post-Early Cretaceous. Restoration of the structures to their pre-rotation position indicates that accretion of the exotic block took place from the northwest and not from the south as previously proposed. This is more consistent with a low latitude origin for the limestone and the general paleogeographic framework. Different tectonic mechanisms and ages for this large rotation are possible. © 2001 Elsevier Science B.V. All rights reserved.

Keywords: paleomagnetism; tectonics; plate rotation; terranes; South America; Southern Andes

1. Introduction

The evolution of the western continental margin of central and southern South America since the Late Paleozoic is an excellent example of a 'pure' subduction regime [1,2], with no evidence of terrane accretion [3]. The only major exception to this is located in southernmost Chile, where massive limestone and basaltic pillow lavas of Late Carboniferous to Early Permian age [4]

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form part of the accretionary prism that extends along the western margin of South America south of 34°S. Tropical fusulinid-bearing limestone at similar paleolatitudes as glacial-derived clastic deposits was taken as evidence of a possible exotic origin for the former by Helwig [5,6]. This author suggested that these carbonate deposits may have been accreted to the continental crust. The faunal assemblages found in the limestone [7] and geochemical data from the pillow lavas [4,7] suggest that these units were originally deposited in an oceanic environment probably far from the present continental margin. The best exposures of these exotic units are in the Madre de Dios Archipelago (around 50°S, 75°W), in southernmost Chile. In order to reconstruct the original paleolatitudes at which these rocks were deposited, a paleomagnetic study was carried out on both the massive limestone (Tarlton Limestone) and the metabasaltic flows (Denaro Complex) within the Madre de Dios Archipelago. Complementary samples were also collected from outcrops at the Diego de Almagro Island (51.5°S, 75°W). Results from this study indicate that most, if not all, the studied rocks have lost their primary remanence, and therefore it is not possible to paleomagnetically reconstruct the original geographic location of these units. However, both units carry a secondary magnetic component at several sites that suggest that the Madre de Dios Archipelago underwent a very large counterclockwise rotation in post-Jurassic times.

A pattern of in situ clockwise rotation of crustal blocks in the Andean arc and fore-arc of Chile has been documented by paleomagnetic studies (see for instance [9–11]), although the timing and cause of these rotations is still controversial. However, most observers agree that a major break in tectonic pattern occurs at about 47–48°S. Sinistral strike-slip faulting in the far south of the south Patagonia batholith has been documented [12]. A large post-batholith counterclockwise rotation of an outboard exotic terrane as suggested by our results tends to confirm a distinctive tectonic pattern for the extreme south of the Chilean margin.

2. Geology and sampling

The most important geological characteristic of the Madre de Dios Archipelago (Fig. 1) is the presence of exotic successions of Late Paleozoic age. According to Forsythe and Mpodozis [4], the stratigraphy of the area comprises four major units: (a) the Denaro Complex, mainly submarine tholeiitic metabasalts, cherts and limestone, interpreted as remnants of paleoceanic floor; (b) the Tarlton Limestone, composed of massive limestone with Late Carboniferous to Early Permian fusulinids and radiolarians [8,13]; (c) the Duque de York complex, made up of flyschoid sediments interpreted as a continent-derived detrital sequence [4] and recently dated as possibly Late Permian [14]; and (4) the south Patagonian batholith, with a granodioritic to tonalitic composition and calc-alkaline affinities [15] that was emplaced into the previous units and has been dated in this area between 132 and 120 Ma approximately [16]. The three former units comprise what is informally called the pre-Late Jurassic basement of the region. They are tectonically disrupted and juxtaposed, leading previous researchers [4] to consider it a 'macromelange'. The degree of deformation varies substantially in short distances in the three units, from virtually undeformed rocks to others with tight ductile deformation [4]. Despite the complex structure of the area, these authors could determine that most thrust planes trend approximately E–W to SE–NW that indicate top to the north transport. They also recognized two phases of deformation, which would pre-date the intrusion of the Early Cretaceous batholith [4].

The same three units of the 'pre-Late Jurassic basement' are exposed 100 km south, in Diego de Almagro Island (Fig. 1) separated by a major shear zone [17] from highly deformed and metamorphosed rocks in the southwestern part of the island.

One hundred and thirty oriented samples were collected with a portable drilling machine at 19 sites: 14 on Madre de Dios and five on Diego de Almagro. Fourteen sites were located in limestone and five in basalts. The distribution of the sampling localities in Madre de Dios is shown in

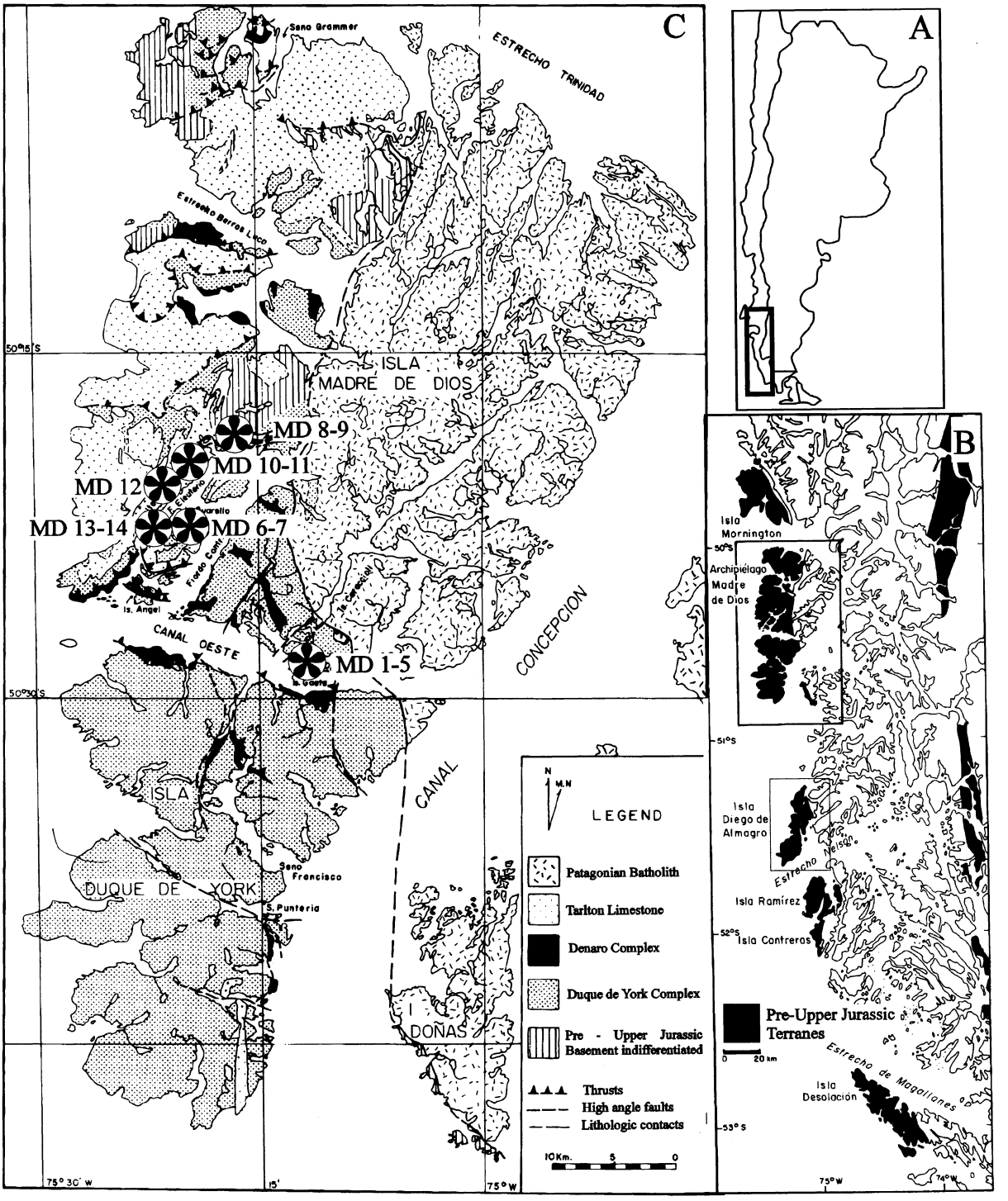


Fig. 1. A: Location map of the study area. B: Distribution of pre-Upper Jurassic rocks in the southwestern margin of South America. Squares show the location of the Madre de Dios and Diego de Almagro islands where paleomagnetic sampling was carried out. C: Geological map of the Madre de Dios Archipelago [4] with location of the sampling localities and site numbers.

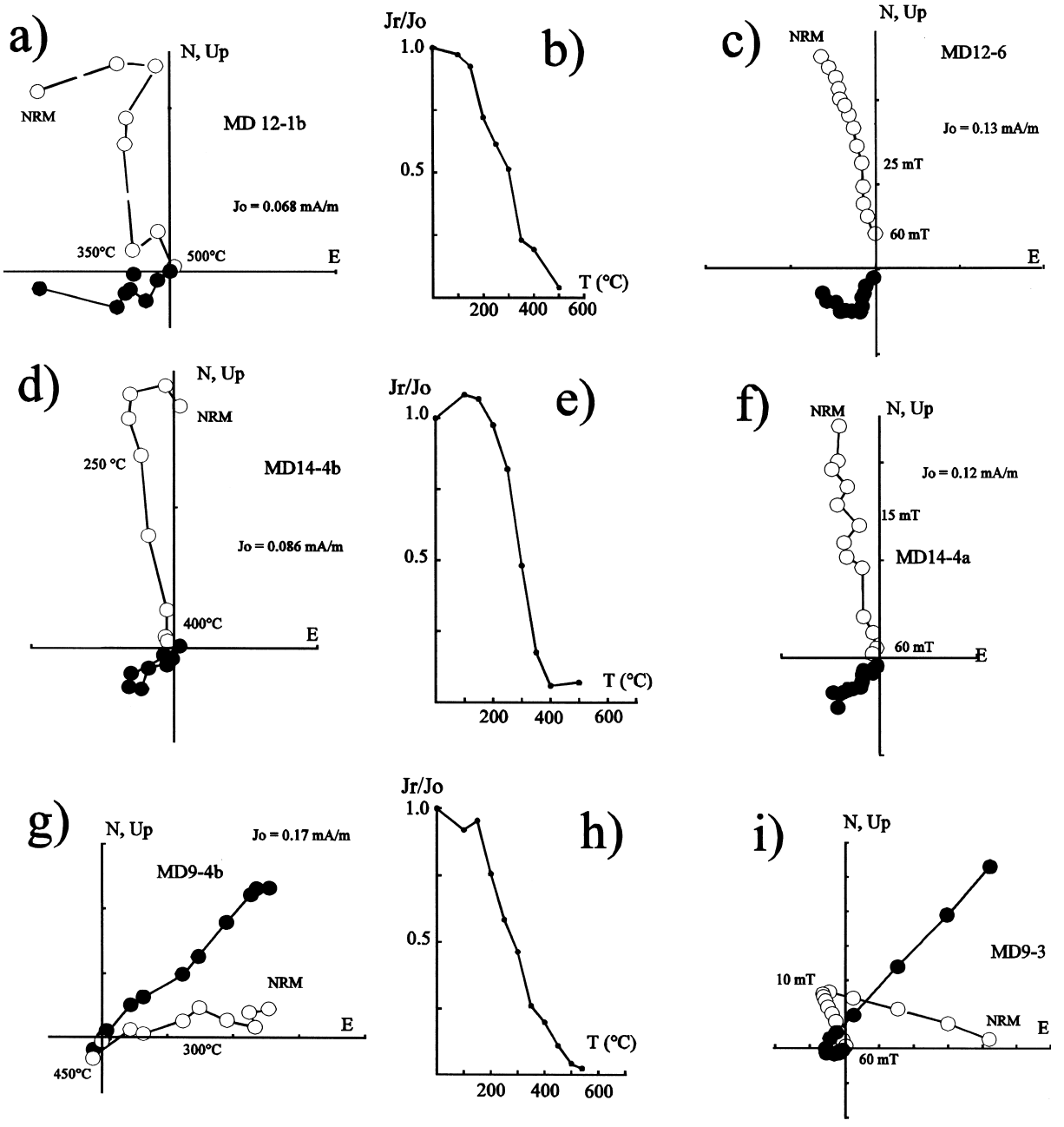


Fig. 2. Demagnetization plots of representative samples from limestone of the Tarlton Formation, subjected to either thermal (a, b, d, e, g and h) or AF cleaning (c, f and i). Representation is by Zijderveld diagrams and normalized intensity decay curves. In Zijderveld plots, full (open) symbols correspond to vector-end projection on the horizontal (vertical) plane. More references in the text.

Fig. 1. Access to the outcrops is severely limited, and could only be done by boat. Weather permitting, samples were oriented with both sun and magnetic compasses and differences were generally kept within 2°. No important orientation errors are therefore expected for those samples oriented solely with magnetic measurements.

3. Paleomagnetic procedures and results

All samples were sliced in the laboratory into one or two standard 2.2 cm high cylindrical specimens (2.54 cm in diameter). Two pilot specimens from a single sample per site were subjected to pilot alternating magnetic fields (AF) or thermal stepwise demagnetization procedures. Up to 19 AF demagnetization steps were applied to peak fields of 160 mT, while 15 thermal steps were made to a maximum temperature of 680°C. According to the magnetic behavior of the pilot specimen, the most suitable procedure to isolate the magnetic components was selected and applied to the remaining specimens (generally one per sample). This was generally the AF method, as the thermal procedure was either unable to discriminate between different components or produced a more erratic or noisy demagnetization behavior (see Figs. 2 and 3). Around 15 steps were applied to each specimen. In a few cases eight to 10 thermal demagnetization steps (up to 625°C) were applied after AF cleaning at 160 mT (Fig. 3g,h). Measurements of intensity and direction of remanence were done with a 2G DC cryogenic magnetometer, AF demagnetization was performed with a three-axis static degausser attached to the magnetometer. Thermal demagnetization was done with a standard Schonstedt TSD-1 oven. During pilot thermal cleaning bulk susceptibility changes were monitored with a susceptibility meter. Zijderveld plots were used for visual inspection of demagnetization behavior and principal component analysis [18] for determination of magnetic components. Not less than four consecutive steps were used to define a magnetic component. Maximum angular deviation (MAD) values only below 15° were accepted, although 77% of components defined had MAD values under 10°. Mean site directions were computed using Fisher's statistics [19].

All sites from Diego de Almagro Island (AL1–AL5) as well as sites MD-6 and MD-7 (Guarello Island quarry) and MD-10 and MD-11 (Seno Eleuterio), in the Tarlton Limestone, presented extremely low natural remanent magnetization (NRM) intensity values (under 10^{-5} A/m) and generally had an erratic demagnetization behavior. Only a few samples from site MD-7 lacked this characteristic but they did not show a consistent direction of magnetization. The remaining sites presented a better magnetic behavior. Their location and lithology are presented in Table 1. Figs. 2 and 3 illustrate several examples of demagnetization behaviors of representative samples from limestone and basalts, respectively.

Samples from the limestone predominantly showed two magnetic components. The low coercivity (<10 mT)/low unblocking temperature (<150°C) component presented a random direction or it was consistent with the expected axial geocentric dipole direction. The high coercivity/high unblocking temperature remanence direction for these sites was generally defined by both AF and thermal treatment (Fig. 2a–f), although thermal demagnetization was sometimes noisier than AF cleaning (Fig. 2a,c). Sometimes, however, thermal cleaning could not discriminate components (Fig. 2g,i). The characteristic remanence (ChRM) presented an in situ west to southwest declination with very high negative inclinations, with the exception of site MD-3 which had shallower inclinations (Table 1). Unblocking temperatures between 400 and 500°C and moderate coercivities suggest a ferrimagnetic mineral as the carrier of the remanence.

On the other hand, the basaltic pillow lavas showed more complex multicomponent demagnetization patterns (Fig. 3). In these cases, thermal demagnetization was not efficient for isolation of the different components (Fig. 3b,e,i). One reason for this failure may be the important chemical changes that many basaltic samples underwent when subjected to experimental high temperatures, as evidenced by changes in bulk susceptibility during thermal demagnetization (Fig. 4A). Two or three magnetic components were found

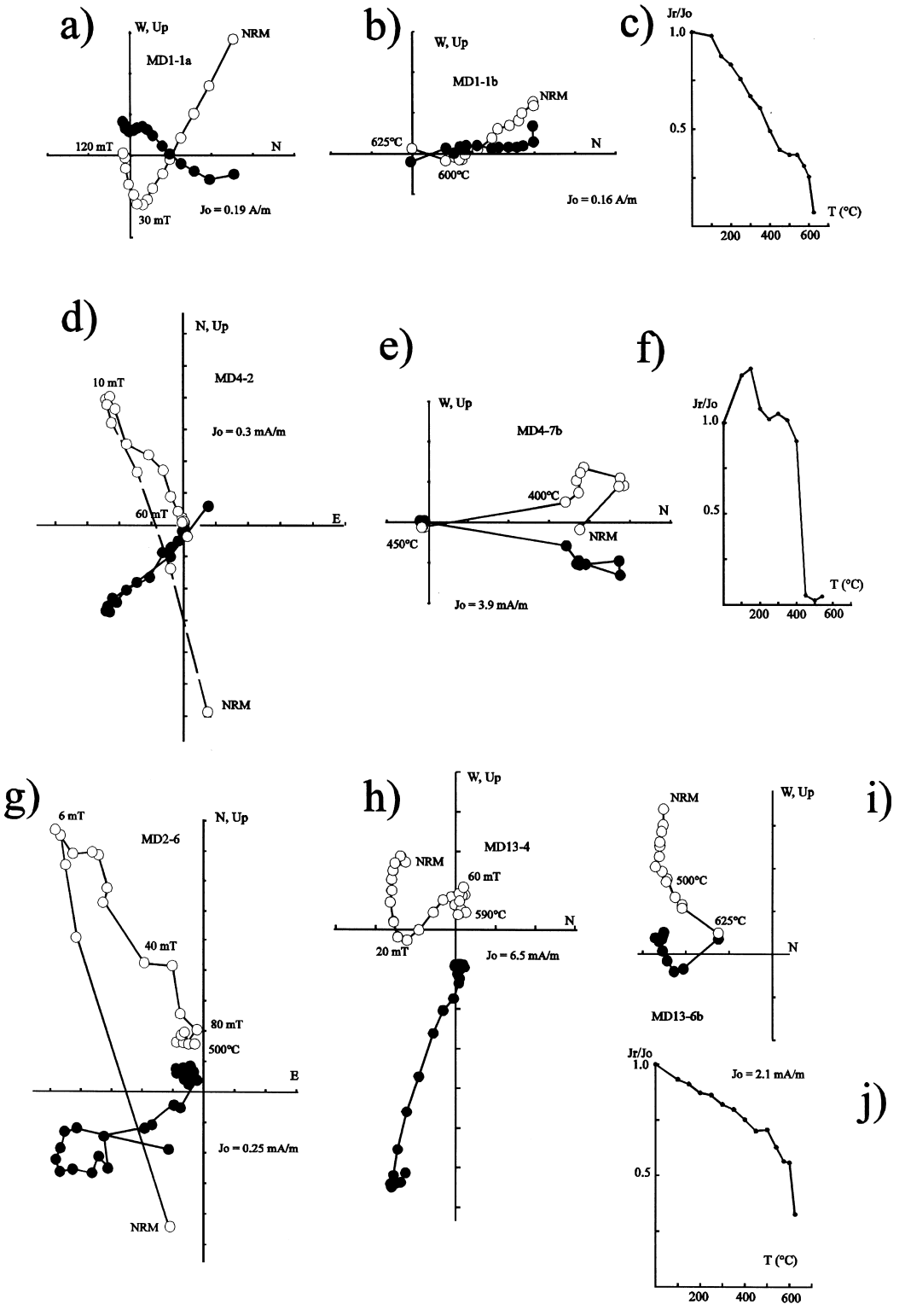

a)
W, Up
NRM
MD1-1a
120 mT
N
30 mT
Jo = 0.19 A/m
b)
W, Up
MD1-1b
NRM
625°C
N
600°C
Jo = 0.16 A/m
c)
Jr/Jo
1.0
0.5
T (°C)
200
400
600
d)
N, Up
MD4-2
10 mT
Jo = 0.3 mA/m
60 mT
E
NRM
e)
W, Up
MD4-7b
400°C
N
450°C
NRM
Jo = 3.9 mA/m
f)
Jr/Jo
1.0
0.5
T (°C)
200
400
600
g)
6 mT
N, Up
MD2-6
40 mT
80 mT
500°C
E
Jo = 0.25 mA/m
NRM
h)
W, Up
MD13-4
NRM
60 mT
590°C
N
20 mT
Jo = 6.5 mA/m
i)
NRM
W, Up
500°C
625°C
N
MD13-6b
j)
Jr/Jo
1.0
Jo = 2.1 mA/m
0.5
T (°C)
200
400
600

Table 1
Mean site directions of the temperature components

Site	Location	Lith.	Geologic unit	*n*	Dec (°)	Inc (°)	α95 (°)	Dec* (°)	Inc* (°)	α95 (°)
A: Low coercivity/unblocking temperature components										
MD-1	I. Gaeta	B	Denaro Complex	6	7.2	−61.6	15.6	300.2	−46.6	
MD-2[a]	I. Gaeta	B	Denaro Complex	7	83.8	69.2	9.7	88.8	23.4	
MD-3	I. Gaeta	L	Denaro Complex	5	349.8	−47.9	33.2	329.4	−36.1	
MD-4[a]	I. Gaeta	B	Denaro Complex	8	70.3	68.6	5.3	84.6	40.0	
MD-8	Seno Eleuterio	L	Tarlton Fm.	4	349.0	−66.3	26.6	79.1	−64.4	
MD-9[b]	Seno Eleuterio	L	Tarlton Fm.	7	47.6	7.4	7.8	40.3	15.1	
MD-12	Seno Eleuterio	L	Tarlton Fm.	5	335.4	65.0	16.0	146.1	−20.8	
MD-13	Sitio IBGT	B	Denaro Complex	8	3.3	−76.4	9.3	135.6	−26.4	
Site mean		L and B	Tarlton and Denaro	5	352.1	−63.9	11.1	115.5	−78.3	72.6
B: Moderate/high coercivity/unblocking temperature components										
MD-1	I. Gaeta	B	Denaro Complex	5	35.6	82.9	10.9	72.7	39.7	
MD-2	I. Gaeta	B	Denaro Complex	8	221.4	−40.3	11.2	235.7	−5.8	
MD-3	I. Gaeta	L	Denaro Complex	5	250.5	−49.3	12.2	258.1	−26.1	
MD-4	I. Gaeta	B	Denaro Complex	6	220.7	−68.4	13.2	251.5	−44.7	
MD-8	Seno Eleuterio	L	Tarlton Fm.	5	276.8	−71.5	15.9	144.6	−69.4	
MD-9	Seno Eleuterio	L	Tarlton Fm.	5	271.2	−77.2	14.8	137.1	−64.2	
MD-12	Seno Eleuterio	L	Tarlton Fm.	8	225.6	−77.9	8.4	160.8	6.7	
MD-13[b]	Sitio IBGT	B	Denaro Complex	7	101.0	−2.4	14.4	74.0	42.8	
MD-14	Sitio IBGT	L	Tarlton Fm	8	216.6	−78.8	3.2	155.9	−12.2	
Site mean		B and L	Tarlton and Denaro	8	237.4	−69.8	12.1	207.4	−43.0	39.6

B: basalts, L: limestone. Dec* and Inc*, declination and inclination values after bedding correction. *n*: number of samples (or sites for computing the mean).
[a]Components interpreted as the same magnetization as those presented in B but with opposite polarity (not considered in the mean).
[b]Component with inconsistent direction (not considered in the mean).

in these rocks. In some cases demagnetization procedures could not delete the remanence completely (Fig. 3a,g–i). A north to northeast upward directed soft component was found in site 1 (Fig. 3a). A south to southwest steeply upward (or north to northeast downward) directed component was found in sites 1, 2 and 4 at fields between 10 and 100 mT. This component is coincident in direction with that determined as the ChRM direction in the limestone. In the latter two sites a nearly antipodal component was determined at lower fields (<15 mT, Figs. 3d,g and 5C). The significance of these opposing components is discussed below. In some cases the remanence was fully demagnetized, but not in others. In the latter cases thermal cleaning was attempted after AF demagnetization up to 140 mT. Since thermal treatment could not progress over 600°C due to chemical changes, it failed to delete the residual remanence. However, it determined an unblocking temperature over 600°C for its carrier (hematite?). Site MD-13 did not show the same components, but rather had a steep upward northerly directed soft component (up to 20 mT or 350°C), coincident with the expected axial geocentric dipole direction, and a nearly horizontal east directed hard magnetization (between 20 and 140 mT). The important presence of an antiferromagnetic fraction in the rocks from this site is evidenced by the impossibility of full demagnetization by AF and unblocking temperatures higher than 600°C (Fig. 3h,i). Whether or not samples

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Fig. 3. Demagnetization plots of representative samples from basaltic pillow lavas of the Denaro Complex. Note that thermal demagnetization is not efficient to separate overlapping magnetic components. References and symbols as in Fig. 2.

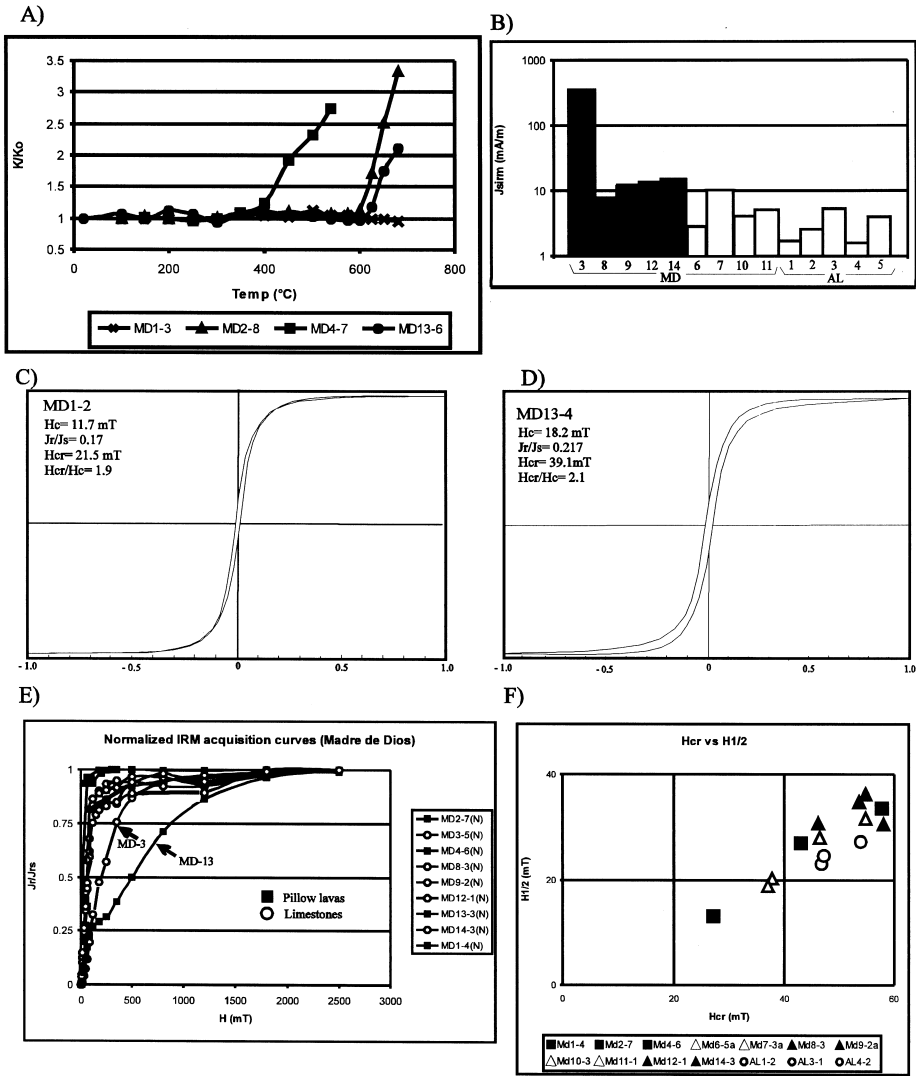


Fig. 4. A: Normalized bulk susceptibility vs. temperature for thermal demagnetization steps of basaltic rocks from the Denaro Complex. Note the two- to three-fold increase in susceptibility at high temperatures. B: Mean site bulk susceptibility for limestone from the Madre de Dios Archipelago (MD) and Diego de Almagro (AL). Numbers refer to sites. In black, sites with a secondary remanent magnetization. In white, sites that presented inconsistent or erratic remanences. C: Hysteresis loop of a sample from the volcanics of site MD-1 with a dominant simple ferrimagnetic signal. D: The same as C but for a sample from site MD-13 with a notorious higher coercivity phase (antiferromagnetic?). E: Isothermal remanent magnetization acquisition curves for sites with stable remanence. Sites MD-13 and MD-3 show an important antiferromagnetic contribution. F: H_{cr} vs. $H_{1/2}$ plot for samples from volcanics of the Denaro Complex (squares), Tarlton limestones with secondary magnetization (full triangles), Tarlton limestones with unstable or erratic remanence (open triangles) and limestones from Diego de Almagro (open circles).

from site MD-13 retain an ancient magnetization is not clear. The possible reason why this site is not carrying the same remanence as the others is discussed below.

Overall, two consistent components were found in the limestone and pillow lavas (Table 1). A soft component consistent with the axial geocentric dipole direction was isolated at sites MD-1, 3, 8, 12 and 13. This is interpreted as a Recent viscous or chemical overprint, which is supported by the fail-

ure of the tilt test ($k_{\text{in situ}}/k_{\text{corrected}} = 22.6$, Table 1) apparently carried by titanomagnetite. A second characteristic component was found in sites MD-1, 2, 4 (basalts), 3, 8, 9, 12 and 14 (limestone) with moderate to high coercivities. This remanence was upward to the west or southwest, except at site MD-1, which carried an antipodal direction. The in situ site mean direction for this component is Dec 237.4°, Inc −69.8°, $\alpha 95$ 12.1°, $n = 8$ sites (Table 1B). The mean site directions for this component are shown in Fig. 5 both in situ and after bedding correction. Visual inspection suggests a post-tectonic remanence. Statistical comparison of k values before and after bedding correction indicates a post-tectonic remanence with 99% significance [20] ($k_{\text{in situ}} = 22$; $k_{\text{corrected}} = 3$). The high dispersion of site remanence directions upon bedding correction cannot be assigned to oversimplified untilting, because dispersion of inclination values also substantially increases with untilting. Since magnetization is post-tectonic, it cannot be older than Late Jurassic, according to the proposed age for the second deformational event that affected the Madre de Dios Archipelago [4,15].

4. Rock magnetism and microscopic observations

In order to better determine the remanence carriers some rock magnetic experiments were carried out. Hysteresis loops were measured with a MicroMag 2900 Alternating Gradient Magnetometer (Princeton Measurements Corp.) on one

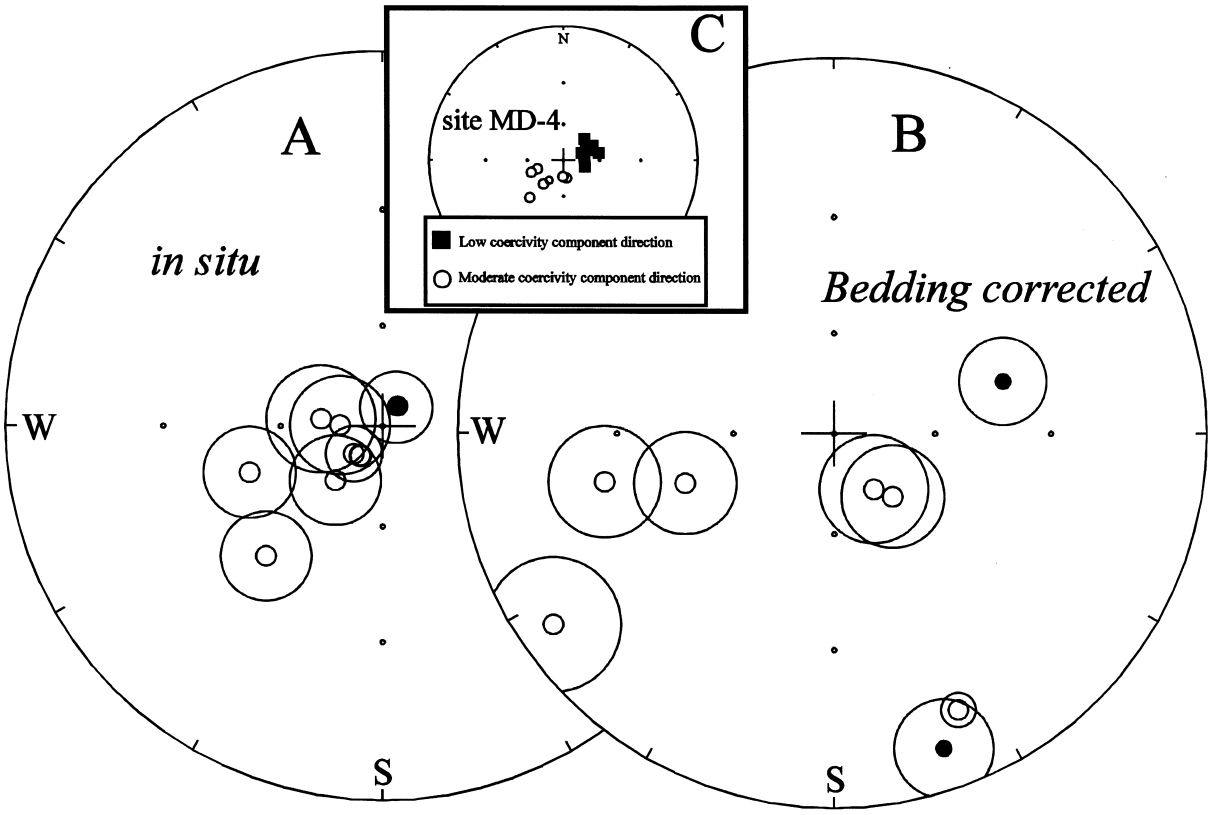


Fig. 5. A: In situ mean site direction for characteristic remanence (moderate to high coercivity/unblocking temperature component) of the Denaro Complex and the Tarlton limestones and their $\alpha 95$. B: The same as A after bedding correction. C: Low coercivity (squares) and moderate coercivity (circles) sample remanence directions from site MD-4. The antipodal directions suggest that different magnetic grains in the samples recorded the same component with opposite polarity.

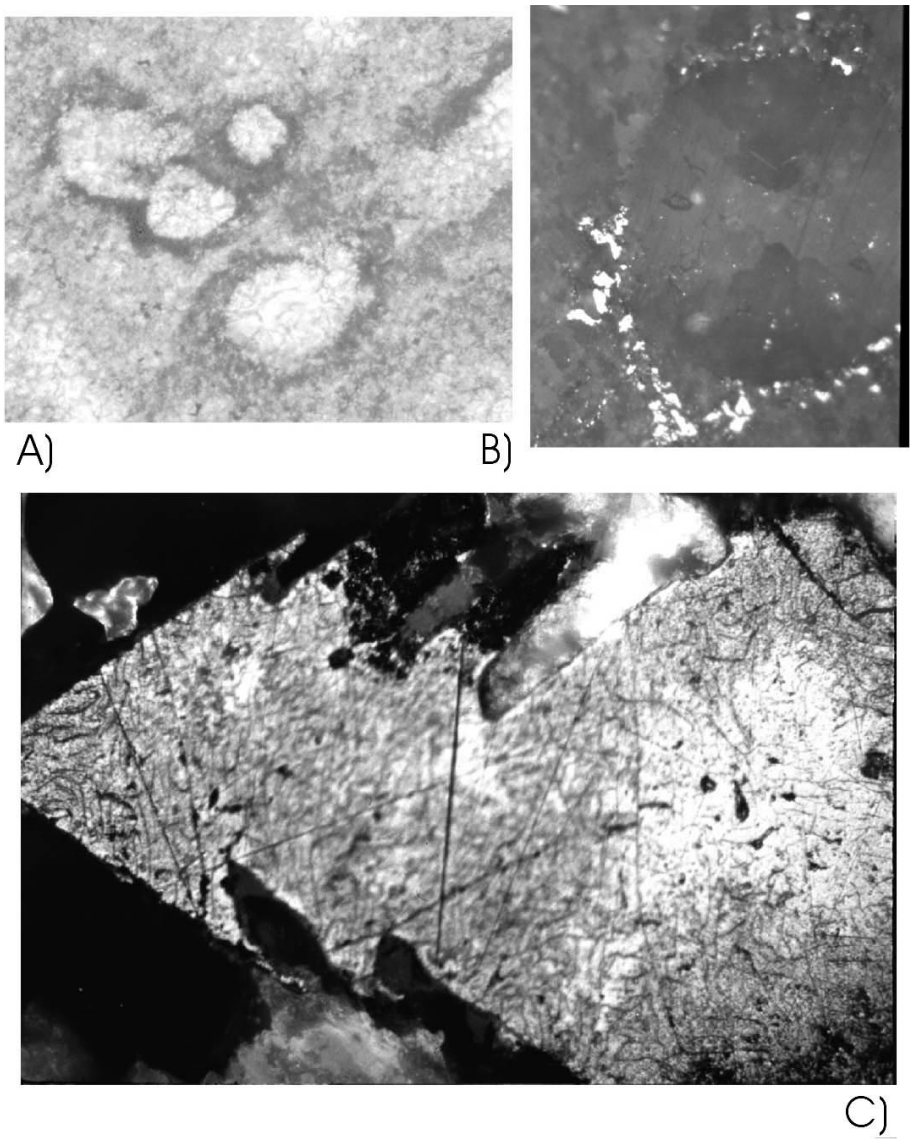


Fig. 6. A: Micrograph of a sample of the Tarlton Limestone. Titanomagnetite grains constitute part of the black rims around the fossils. Transmitted light. B: Titanomagnetite grains (~4–6 μm in size) rimming a fossil. Reflected light, in oil. C: Maghemitization and surface cracking in a titanomagnetite crystal (~260 μm in size) from a pillow lava of the Denaro Complex. Brighter areas in the crystal correspond to less oxidized titanomagnetite. Reflected light, in oil.

sample from each site of the basaltic flows (Fig. 4c,d). Sample MD-1-2 shows a typical loop for a single ferrimagnetic fraction, with values of hysteresis parameters that suggest pseudo-single domain (PSD) magnetite. On the other hand, sample MD-13-4 shows the presence of a higher coercivity fraction, with saturation reached only near 1 T. Evidently both ferrimagnetic and antiferromagnetic phases are present. This is consistent with the NRM demagnetization behavior of samples from these sites.

Isothermal remanent magnetization (IRM) acquisition curves were obtained for at least one sample per site (Fig. 4e) with a pulse magnetizer.

These curves suggest that a ferrimagnetic fraction is the dominant carrier in most limestones and basalts. The main exceptions are sites MD-3 (limestone from the Denaro Group) and MD-13 (basalt), in which the presence of an antiferromagnetic fraction is very conspicuous. Backfield IRM was applied to all these samples in order to determine their remanent coercive forces (H_{cr}). Their saturation remanent magnetization was also demagnetized by a stepwise AF procedure. This permitted the calculation of their medium destructive fields ($H_{1/2}$). For comparison, these values were also obtained from sites at Madre de Dios with no stable remanence and those from Diego de Almagro. A H_{cr} vs. $H_{1/2}$ plot is presented in Fig. 4f. Samples with an important antiferromagnetic contribution were excluded. While the basalts show a broad range of H_{cr} values, the limestones tend to be better grouped, with those from Madre de Dios with a stable remanence (full triangles in Fig. 4f) showing the higher H_{cr} values and $H_{1/2}/H_{cr}$ ratio. In most cases, H_{cr} values are consistent with single domain or PSD magnetite [21,22]. Important differences between values of H_{cr} and $H_{1/2}$ for all samples may suggest magnetic interaction between grains [21], although this effect can also be caused by multidomain grains [23]. This difference is higher in the samples from Diego de Almagro and Madre de Dios with no stable remanence. There is another systematic difference in rock magnetic parameters between those samples with stable remanence and those without it. Fig. 7 shows total intensity values of saturation remanence for different limestone sites. All samples that carry a stable remanence show systematically higher values. These elements suggest that a secondary ferrimagnetic fraction (titanomagnetite?) is present in those limestone sites from which the post-folding ChRM was computed. MD-3 shows an exceptionally high value, which may be due to a difference in lithology content (this is a 10 cm thin bed intercalated in the volcanic succession and the only sampled limestone belonging to the Denaro Com-

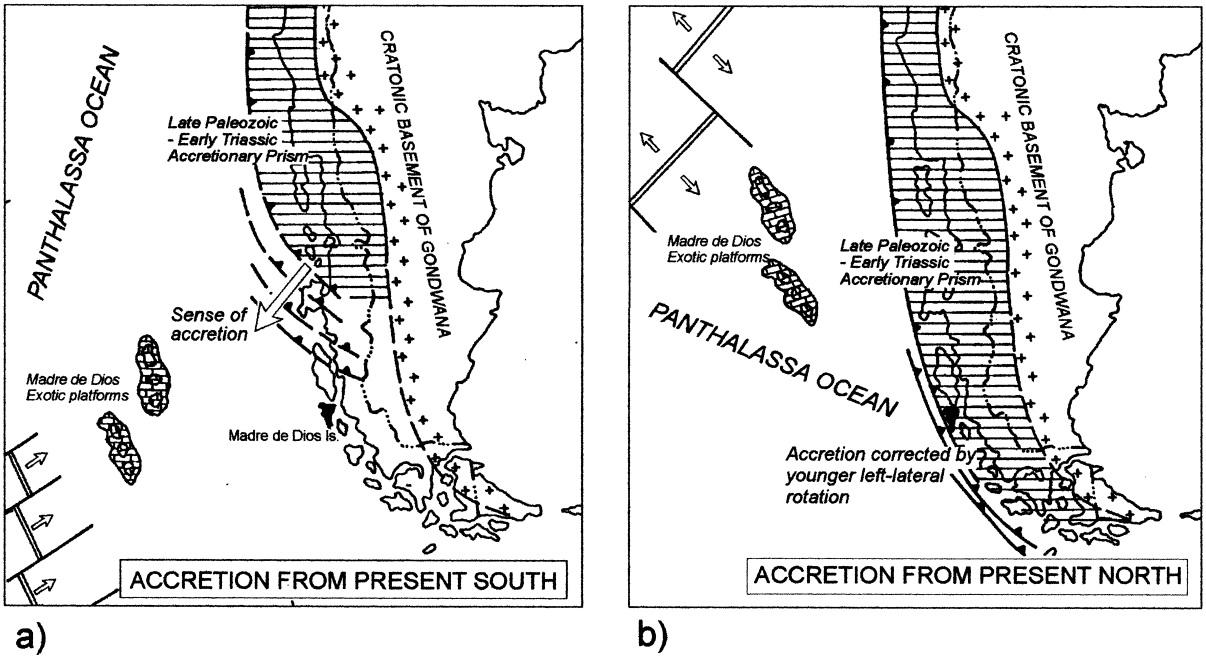


Fig. 7. Paleogeographic sketches depicting docking and amalgamation of the Madre de Dios carbonate platforms to the margin. a: Previous proposals that accepted the present structural trend of Madre de Dios, and suggested a derivation from the south [2,15,29]. b: Alternative interpretation after restoration of structures for a large counterclockwise rotation, where docking has a north source in agreement with early proposals of Helwig [5,6] and Monger [31].

plex), although it could also be related to its proximity to the batholith (see below).

Thin and polished sections of the studied rocks were microscopically analyzed by transmitted and reflected light in order to recognize the magnetic minerals and their paragenesis. The limestone shows a varied degree of recrystallization [15] with abundant microfossils generally replaced by carbonates (Fig. 6A). Opaque minerals are scarce. Titanomagnetite is the main magnetic mineral, occurring as small crystals (generally <10 μm) in fossil rims (Fig. 6A,B), microfractures and microstylolites. Pyrite framboids were also identified in several samples. The microscopic findings are consistent with the rock magnetic data, which suggest titanomagnetite as the likely carrier of the remanence. The distribution of the titanomagnetite crystals also suggests a secondary origin consistent with the paleomagnetic results.

The pillow lavas also show a varied degree of alteration with evidence of a low-grade metamorphism (greenschist), consistent with their postulated origin as ocean floor basalts. The main magnetic mineral is partially oxidized titanomagnetite (titano-maghemites) with hematite as a subordinated mineral. Titanomagnetites occur as large skeletal crystals (in many cases over 100 μm, Fig. 6C) with a trend towards a euhedral morphology, likely as a result of rapid crystallization. The maghemitization is not uniform throughout the grains and the typical curved cracks are observed (Fig. 6C). Hematite is more abundant in site MD-13 consistent with the rock magnetic data. Pyrite is also an abundant opaque mineral.

5. Discussion and interpretations

From the paleomagnetic results it can be inferred that most studied rocks do not carry a primary magnetic remanence. Therefore, the main aim of the study, i.e. to determine the original paleogeographic position of the exotic rocks of Madre de Dios Archipelago, cannot be accomplished. However, since a consistent characteristic remanence was determined in several sites, its origin and tectonic implications will be examined.

From the paleomagnetic analysis, rock magnetic experiments and microscopic observations, it is very likely that the post-tectonic characteristic remanence determined in the limestone is carried by secondary titanomagnetite deposited in microfractures, stylolite and fossil rims. The carrier of this remanence in the basalts is somewhat more problematic as the oxidized titanomagnetite crystals are apparently of primary crystallization and maghemitization is a common early process in seafloor basalts. The most likely source for the remagnetization found in these rocks is the intrusion of the south Patagonian batholith ca. 130 Myr ago [16] (see Fig. 1). Heat transfer and hydrothermal activities associated with it may have produced titanomagnetite precipitation in the limestone. This is supported by the fact that a magnetite mineralization has been reported along the contact between the intrusive and the Tarlton Limestone in the Dueñas island, some 30 km south of Madre de Dios [15]. It is also interesting to note that the remagnetization of the pillow lavas was only found at those sites close to the intrusion (i.e. <5 km). There is also a markedly higher magnetic mineral content in the limestone of site MD-3 than the others, which may suggest a decrease in titanomagnetite production with distance to the intrusion. However, the lithology of site MD-3 (thin layer of limestone interbedded in the Denaro Complex, deposited in the vicinity of volcanic and hydrothermal activity) is not strictly comparable with that from the Tarlton Limestone. Opposite magnetic polarities were recorded at different sites (Fig. 5A,B); and two antipodal components were found at sites MD-2 and MD-4 (Fig. 5C) carried by magnetic phases with different coercive forces (Fig. 3d,g). A simple explanation for this is to consider that the same magnetization was recorded with dual polarities at these sites. All this suggests that remagnetization took place during an interval in which at least one polarity reversal of the Earth's magnetic field occurred. This is consistent with remagnetization occurring at ca. 130 Myr ago and makes less likely its occurrence during the Cretaceous Normal Superchron (118–84 Ma [24]).

In situ mean site directions (Fig. 5A) show an anomalous declination. Values of rotation and paleolatitude anomaly were computed [25,26] for

the Madre de Dios Archipelago from the overall site mean direction. The reference direction used was that from the Early Cretaceous Serra Geral pole [27] as recalculated by Beck [28]. The values obtained are R (rotation): $117.7 \pm 29.9°$ (counterclockwise), F (inclination anomaly): $5.4 \pm 9.7°$. Using reference poles of different ages does not significantly change the results.

A large rotation around a vertical axis close to the sampling localities can explain the paleomagnetic results. Although the uncertainties are large, the results indicate a very large counterclockwise rotation after acquisition of the remanence. The anomaly in paleolatitude is not significant and indicates that coastwise transport of the rocks has been negligible.

Distribution of the sites that contribute to the mean direction implies that at least an area of over 400 km^2 was affected by rotation. The fact that sites with significantly different attitudes show similar declination values strongly suggests a rigid body rotation that postdates deformation of the rocks. Despite many local deflections, a main systematic structural trend has been recognized in the archipelago [4], with no reported significant changes at different localities. It is likely then, but not proved, that the rotated area encompasses most if not all outcrops of the pre-Jurassic basement of Madre de Dios. The structural trends must therefore be rotated by some 90–145° clockwise to reconstruct their position prior to tectonic rotation. This leads to an approximate northeast trend for the thrusts, implying a NW–SE (present coordinates) transport direction. This is consistent with these structures being related to the accretionary process of these exotic units, as it would imply an approximation direction and collision with the paleocontinental margin from the northwest instead of from the south (Fig. 7). The latter, proposed by previous authors [2,15,29] on the basis of the present structural trend, would be an unexpected direction of arrival for these exotic units, as it is difficult to imagine a thick carbonate platform developing in very high latitudes. Considering that South America has not dramatically changed its position with respect to the South Pole since the Late Permian [30], restoration of the Madre de Dios rocks to their pre-rotation position suggests a provenance from the north to northwest in a Late Paleozoic–Early Mesozoic paleogeographic framework. In an analogous situation, Monger [31] recognized similar fussulinid-bearing carbonates capping oceanic tholeiites in the intermontane belt of British Columbia. Based on paleomagnetic data and paleoclimatic considerations this author claimed that the Cache Creek terrane was derived from subequatorial latitudes. Similar considerations can be imposed for the Madre de Dios carbonates.

An alternative interpretation to the large counterclockwise rotation is to rotate the mean remanence direction obtained in our study 40° around a horizontal axis of azimuth 300° (clockwise looking towards the northwest). This would restore our anomalous direction to the expected one and would imply that after acquisition of the remanence the sampling area experienced a 40° southward tilt around a horizontal axis. This is considered less likely than the rotation around a vertical axis for the following reasons: (i) since no major post-deformational east to northeast faults, either normal or reverse, have been reported in the archipelago, and similar structural vergence is seen on most outcrops [4], tilting would probably have to encompass most of or the whole archipelago, which would imply a difference in structural level for different sampling sites of over 7 km and over 30 km for outcrops of the same units in the area; no evidence has been reported for this; (ii) a 40° tilt would not substantially change the structural trends, but most folds and faults would invert their vergence to the north to northeast; this would imply a NE–SW transport direction which would be very difficult to account for, given the position of the terrane on the southwestern margin of the continent; (iii) some of the structures, however, would still remain with a steep south to southwest vergence or even be vertical, substantially changing the structural pattern of the region; (iv) similar inclination values between the calculated mean remanence direction and the reference one should be assigned to coincidence.

Therefore, the interpretation of a large counterclockwise rotation is considered the simplest way to account for the paleomagnetic data.

If the remagnetization found in Madre de Dios is due to the intrusion of the Early Cretaceous batholith, as proposed here, then the age of rotation is only constrained to be post-Early Cretaceous. Different tectonic processes could have produced this rotation. They are briefly analyzed below.

5.1. Late Cenozoic rotation

The western continental margin of southern South America shows major strike-slip faults parallel to the margin, the most prominent being the Liquiñe–Ofqui fault zone [32], between around 39 and 47°S. South of Golfo de Penas, the existence of northwest trending left-lateral strike-slip faults have been inferred [33]. Earthquake focal mechanisms indicate left-lateral motion for a northwest trending fault (Fig. 8c). The Madre de Dios Archipelago is located between two of these major faults (Fig. 8c). Displacement along them could produce counterclockwise rotation of a crustal

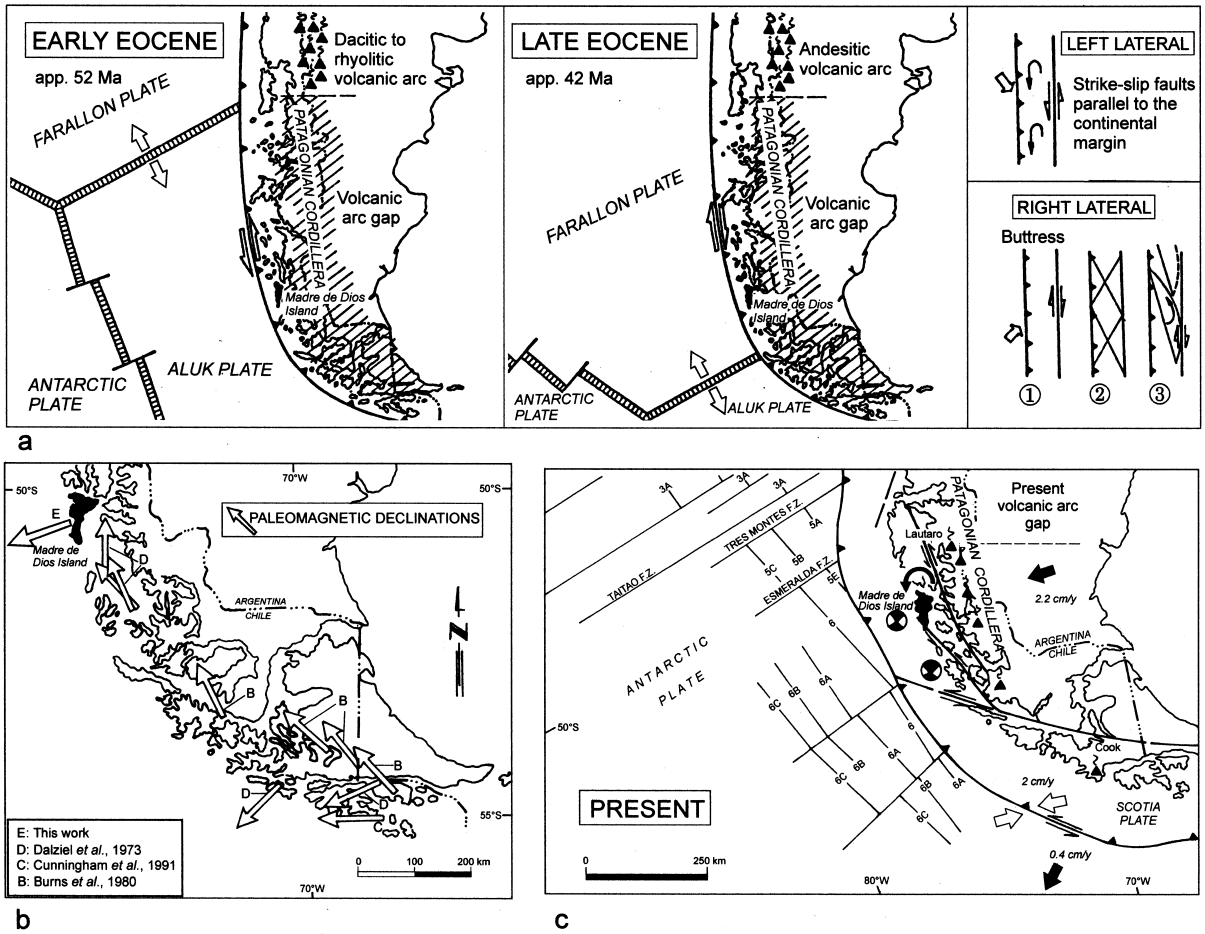


Fig. 8. Speculative tectonic processes that may have caused the large counterclockwise rotation determined for the Madre de Dios Archipelago. a: Highly oblique subduction in Paleogenic times may have caused the rotation. This could happen either in a left-lateral (Early Eocene) or in a right-lateral shear zone (Late Eocene). Schematic models for rotation after Rojas et al. [39] are presented. b: Declination anomalies related to oroclinal bending of southernmost Andes found by previous studies (B, C and D) and this work (E). c: Present tectonic setting active since Late Cenozoic times. Note that the study area is located between two active left-lateral strike-slip faults. Earthquake focal mechanisms indicate left-lateral motion for a northwest trending fault, after Fuenzalida [43] and Ramos and Kay [42].

block comprising the archipelago. However, the amount of displacement along these faults is not known (C. Mpodozis, oral communication). These strike-slip faults could be related to final stages of development of the Patagonian or Magellanes oroclines [12]. The curvature of the southern tip of South America has been interpreted by some as an orocline developed in Cretaceous to Cenozoic times [12]. Scarce paleomagnetic data on rocks along the orocline show almost systematic counterclockwise rotations that increase southward and that follow approximately the trend of the continental margin (Fig. 8b [34–36]). These rotations reach over 90° in southern Tierra del Fuego [36]. Madre de Dios Archipelago is located close to the northern limit of the proposed orocline and therefore should not be affected by these rotations. However, Cunningham [12] has presented a model in which the orocline developed by a series of left-lateral wrench faults that migrated northward. The northwest trending faults mentioned above may have developed as a result of this process. This would suggest a Late Cenozoic rotation.

5.2. Early Cenozoic rotation

Rotation could also have occurred associated with oblique subduction along the continental margin of southern South America. Highly oblique subduction developed along this margin in Paleogenic times [37], dextral between 42 and 25 Ma (Farallones plate) and sinistral between 52 and 42 Ma (Aluk plate, see Fig. 7a). Large in situ rotation of fore-arc slivers is common in oblique subduction regimes [38]. Although a counterclockwise rotation of a fore-arc sliver is easily accommodated in a left-lateral oblique subduction regime, it can also occur under specific circumstances in a right-lateral one [39].

5.3. Cretaceous rotation

In Late Jurassic–Early Cretaceous times, a marginal basin developed in southern South America to the east of the magmatic arc. It is known as the Rocas Verdes or Magellanes marginal basin and it consisted of an oceanic-floored back-arc basin. It extended from about 49 to 56°S. Its closure, mainly in mid-Cretaceous times, produced pervasive ductile deformation [2] along the foreland margin of the basin. Recent structural models [40,41] have emphasized the importance of strike-slip displacements during this process. A large rotation of the Madre de Dios area within this context cannot be ruled out.

The speculations made above do not include tectonic processes associated with ridge collisions. The Chile ridge collided with the continental margin at the latitude of Madre de Dios Archipelago some 10 Myr ago [37]. Previous to this, other ocean ridge collisions occurred [42]. The relative importance of such collisions in promoting large crustal block rotations remains to be examined.

Many of the above-mentioned possible scenarios could be ruled out if the age of the rotation could be determined. Paleomagnetic data from the south Patagonian batholith may shed some light on these issues, however, lack of exposures of Cenozoic rocks would certainly severely restrict the paleomagnetic possibilities.

6. Conclusions

A paleomagnetic study carried out on the Late Carboniferous–Early Permian exotic Tarlton Limestone and Denaro Complex, exposed in the Madre de Dios Archipelago and Diego de Almagro Island, in southern Chile, failed to recover the original remanence of these rocks. However, a post-tectonic characteristic remanence was determined in eight sites, both in limestone and in basaltic pillow lavas. Rock magnetic experiments and microscopic observations determined that the carrier of this remanence was secondary titanomagnetite in the limestone and titanomagnetite–titanomaghemite in the lavas. The most likely event to produce this secondary magnetization was the intrusion of the south Patagonian batholith in the Early Cretaceous. Site mean direction disagrees with the reference direction. It is considered that the simplest way to account for the paleomagnetic discrepancy is a very large ($117.7 \pm 29.9°$) counterclockwise rotation of the study area with negligible paleolatitude anomaly.

This rotation is only constrained as post-Early Cretaceous. Restoration of the structures to their pre-rotation position suggests that accretion of the exotic block took place from the northwest, and not from the south to southwest as previously thought. This is consistent with these rocks being formed at lower latitudes. Different tectonic mechanisms and ages for this large rotation are possible.

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References

[1] I.W. Dalziel, R.D. Forsythe, Andean evolution and the terrane concept, tectonostratigraphic terranes of the circum-Pacific region, Earth Sci. Ser. 1 (1985) 565–581.

[2] C. Mpodozis, V.A. Ramos, The Andes of Chile and Argentina, in: G.E. Ericksen, M.T. Cañas Pinochet, J.A. Relnemund (Eds.), Geology of the Andes and its Relation to Hydrocarbon and Mineral Resources, Circum-Pacific Council for Energy and Mineral Resources, Houston, TX, Earth Sci. Ser. 11 (1989) 59–90.

[3] M.E. Beck Jr., R.R. Burmester, R.E. Drake, P.D. Riley, A tale of two continents: Some tectonic contrasts between the central Andes and the North American Cordillera, as illustrated by their paleomagnetic signatures, Tectonics 13 (1994) 215–224.

[4] R.D. Forsythe, C. Mpodozis, El archipielago Madre de Dios, Patagonia Occidental, Magallanes rasgos generales de la estratigrafía y estructura del 'basamento'pre-Jurásico Superior, Rev. Geol. Chile 7 (1979) 13–29.

[5] J. Helwig, Stratigraphy, sedimentation, paleogeography and paleoclimates of Carboniferous ('Gondwana') and Permian of Bolivia, Am. Assoc. Petrol. Geol. Bull. 56 (1972) 108–133.

[6] J. Helwig, Late Paleozoic and stratigraphy of the Central Andes, An. Acad. Brasil. Ci. 44 (Suppl.) (1973) 161–172.

[7] R.C. Douglass, M.K. Nestell, Late Paleozoic foraminifera from southern Chile, U.S. Geol. Survey, Special Paper 858 (1976) 49 pp.

[8] H.Y. Ling, R.D. Forsythe, C.R. Douglass, Late Paleozoic microfaunas from southernmost Chile and their relation to Gondwanaland forearc development, Geology 13 (1987) 357–360.

[9] R. Somoza, S. Singer, B. Coira, Paleomagnetism of upper Miocene ignimbrites at the Puna: an analysis of vertical-axis rotations in the Central Andes, J. Geophys. Res. 101 (1996) 11389–11400.

[10] M.E. Beck Jr., On the mechanism of crustal block rotations in the central Andes, Tectonophysics 299 (1998) 75–92.

[11] D.E. Randall, A new Jurassic–Recent apparent polar wander path for South America and a review of central Andean tectonic models, Tectonophysics 299 (1998) 49–74.

[12] W.D. Cunningham, Strike-slip faults in the southernmost Andes and the development of the Patagonian orocline, Tectonics 12 (1993) 169–186.

[13] H.Y. Ling, R.D. Forsythe, Late Paleozoic pseudoalbaillellid radiolarians from southernmost Chile and their geologic significance, in: D.H. Elliot, J.W. Collinson, G.D. Mckenzie (Eds.), Gondwana Six, Am. Geophys. Union, Geophys. Monogr. Ser., 1987, pp. 253–260.

[14] F. Hervé, M. Faning, J. Bradshaw, M. Bradshaw, J. Lacassie, Late Permian shrimp U-Pb detrital zircon ages constrain the age of accretion of oceanic basalts to the Gondwana margin at the Madre de Dios archipelago, southern Chile, 4th Int. Symp. on Andean Geodynamics (ISAG), Göttingen, Extended Abstracts, 1999, pp. 327–328.

[15] R. Forsythe, C. Mpodozis, Geología del basamento pre-Jurásico Superior en el archipélago Madre de Dios, Magallanes, Chile, Servicio Nacional de Geología y Minería, Chile, Bol. 39 (1983) 63 pp.

[16] M. Halpern, Regional geochronology of Chile south of 50° latitude, Geol. Soc. Am. Bull. 84 (1973) 2407–2422.

[17] R. Forsythe, I.W. Dalziel, C. Mpodozis, B. Hall, S. Barrientos, Geologic studies in the outer Chilean fjords, R.V. Hero cruise 79-5, Antarct. J. USA 15 (1981) 109–111.

[18] J.L. Kirshvink, The least-squares line and plane and the analysis of palaeomagnetic data, Geophys. J. R. Astron. Soc. 62 (1981) 699–718.

[19] R.A. Fisher, Dispersion on a sphere, Proc. R. Soc. London A 217 (1953) 295–306.

[20] M.W. McElhinny, Statistical significance of the fold test

in palaeomagnetism, Geophys. J. R. Astron. Soc. 8 (1964) 338–340.
[21] S. Cisowski, Interacting vs non-interacting single domain behaviour in natural and synthetic samples, Phys. Earth Planet. Interact. 26 (1981) 56–62.
[22] P. Dankers, Relationship between median destructive field and remanent coercive forces for dispersed natural magnetite, titanomagnetite and hematite, Geophys. J. R. Astron. Soc. 64 (1981) 447–461.
[23] D.J. Dunlop, Ö. Özdemir, Rock Magnetism, Fundamentals and Frontiers, Cambridge University Press, Cambridge, 1997, 573 pp.
[24] J.G. Ogg, Magnetic polarity time scale of the Phanerozoic, in: T.J. Ahrens (Ed.), Global Earth Physics, A Handbook of Physical Constants, AGU Reference Shelf 1 (1995) 240–270.
[25] H.H. Demarest Jr., Error analysis for the determination of tectonic rotation from paleomagnetic data, J. Geophys. Res. 88 (1983) 4321–4328.
[26] M.E. Beck Jr., Paleomagnetism of continental North America; Implications for displacement of crustal blocks within the Western Cordillera, Baja California to British Columbia, Geol. Soc. Am. Mem. 172 (22) (1989) 471–492.
[27] M. Ernesto, I. Pacca, F. Hiodo, A. Nardy, Paleomagnetism of the Serra Geral Formation, southern Brazil, Phys. Earth Planet. Interact. 64 (1990) 153–175.
[28] M.E. Beck Jr., On the mechanism of crustal block rotations in the central Andes, Tectonophysics 299 (1998) 75–92.
[29] C. Mpodozis, R. Forsythe, Stratigraphy and geochemistry of accreted fragments of the ancestral Pacific floor in southern South America, Paleogeogr. Paleoclimat. Paleoecon. 41 (1983) 103–124.
[30] J. Golonka, M.I. Ross, C.R. Scotese, Phanerozoic paleogeographic and paleoclimatic modeling maps, Can. Soc. Petrol. Geol. Mem. 17 (1993) 1–47.
[31] J.W.H. Monger, Cordilleran tectonics: a Canadian perspective, Bull. Soc. Geol. Fr. 7 (1984) 255–278.
[32] J. Cembrano, F. Hervé, A. Lavenu, The Liquiñe-Ofqui fault zone a long-lived intra-arc fault system in southern Chile, Tectonophysics 259 (1996) 55–66.
[33] J. Corvalán, Paleotectonic Map of the Circumpacific region, Southeast Quadrant, Circum-Pacific Council for Energy and Mineral Resources, A.A.P.G., Scale 1:10,000,000, 1981.
[34] I.W.D. Dalziel, R. Kligfield, W. Lowrie, N.D. Opdyke, Paleomagnetic data from the southernmost Andes and the Antarctandes, in: D.H. Tarling, S.K. Runcorn (Eds.), Implications of Continental Drift to Earth Sciences, Vol. 1, Academic Press, San Diego, CA, 1973, pp. 87–101.
[35] K.L. Burns, M.J. Rickard, L. Belbin, F. Chamalaun, Further paleomagnetic confirmation of the Magellanes orocline, Tectonophysics 63 (1980) 75–90.
[36] W.D. Cunningham, K.A. Klepeis, W.A. Gose, I.W. Dalziel, The Patagonian Orocline: New paleomagnetic data from the Andean magmatic arc in Tierra del Fuego, Chile, J. Geophys. Res. 96 (1991) 16061–16069.
[37] S. Cande, R.B. Leslie, Late Cenozoic tectonics of the southern Chile trench, J. Geophys. Res. 91 (1986) 471–496.
[38] M.E. Beck, Jr., Block rotations in continental crust: examples from Western North America, in: C. Kissel, C. Laj (Eds.), Paleomagnetic Rotations and Continental Deformation, Kluwer Academic, Dordrecht, 1988.
[39] C. Rojas, M.E. Beck Jr., R.F. Burmester, J. Cembrano, F. Hervé, Paleomagnetism of the mid-Tertiary Ayacara Formation, southern Chile: counterclockwise rotation in a dextral shear zone, J. S. Am. Earth Sci. 7 (1994) 45–56.
[40] W.D. Cunningham, Orogenesis at the southern tip of the Americas: the structural evolution of the Cordillera Darwin metamorphic complex, southernmost Chile, Tectonophysics 244 (1995) 197–229.
[41] P.E. Kraemer, Palinpastic restoration of the Patagonian Orocline (50°S–56°S): A first attempt, 14° Congr. Geol. Argent. 1 (1999) 241–244.
[42] V.A. Ramos, S. Mahlburg Kay, Southern Patagonia plateau basalts and deformation: backarc testimony of ridge collisions, Tectonophysics 205 (1992) 261–282.
[43] P.R. Fuenzalida, The Magellan Fault Zone, in: O. Gonzalez Ferrán (Ed.), Symposium on Andean and Antarctic Volcanology Problems, Int. Assoc. Chem. Earth Inter., Special Series, Naples, 1976, pp. 373–391.